

Components of an Industrial Control System and SCADA Network



- **Industrial System:**

- It is a group of machines working together, such as in an assembly line.
- Computers work with physical components that are not necessarily digital.
- These systems can cover large areas, like transportation networks or gas pipelines.

- **Internet of Things (IoT):**

- Some people see IoT as the next step in internet development.
- It connects everyday items that are not normally used as computers.

- **Software and ICS servers** are key parts of an Industrial Control System (ICS).

- **Acquisitions server (I/O server):** Collects input/output data from devices.
- **Control server / MTU / SCADA server:** Manages system operations and communications.
- **Historian:** Stores historical data for analysis and reporting.



SCADA (Supervisory Control and Data Acquisition):

A system used to monitor and control industrial processes.

- Collects real-time data from sensors and machines.
- Helps operators supervise and manage complex systems remotely.
- Used in places like:
 - Power plants
 - Water treatment facilities
 - Manufacturing and production lines

- **Two control methods in Industrial Control Systems (ICS):**

- **Open loop system:**

- Makes decisions using predefined rules, past events, and history.

- **Closed loop system:**

- Makes decisions using real-time data, adjusts the system dynamically
- Needs field devices placed across the system to monitor operations (known as a Distributed Control System – DCS).

Asset management

- **Asset management documentation:**

- It is important when managing a large number of devices.
- It is essential for organizing systems in enterprise environments.

- Asset management means tracking and maintaining all components in a network.

- The first step is to list all assets:

- Hardware devices (nodes)
- Software

- Organizations must choose the right asset management tool based on their needs.

- **Benefits:**

- Makes it easier to maintain and upgrade the network.
- Gives details about costs and advantages of hardware and software.

! Business documents

Common types of business documents:

- **RFP (Request for Proposal):**
 - A request sent to vendors to propose a product or service the company wants to buy.
- **MOU (Memorandum of Understanding):**
 - Shows the intent of two or more parties to form a formal agreement or contract.
- **SOW (Statement of Work):**
 - Describes the specific tasks and goals that need to be done for a project.
- **SLA (Service-Level Agreement):**
 - A legal agreement that defines the details of a service given to a customer.
 - Example: Internet service provided by an ISP.
- **MLA (Master License Agreement):**
 - Gives permission from the creator or developer to another party for selling, sublicensing, or distributing the product.

!! Software and Hardware Changes

There are three types of changes made to software:

- **Patch:**
 - A fix, improvement, or small update to correct issues.
- **Upgrade:**
 - A big change that adds new features or improves existing ones.
- **Rollback:**
 - Also known as backleveling or downgrading.
 - Reverting to a previous version after a patch or upgrade fails.

General steps to change software or hardware:

- Don't let patches install automatically.
- Check if a patch or upgrade is really needed.
- Research why the change is needed, and check compatibility and effects.
- Test the patch or upgrade in a lab to confirm it works properly.
- Decide if the change should affect all users, parts of the network, or specific devices.
- Plan the update during off-hours (called a **maintenance window**)
- Notify the right team members about the planned change.
- Back up the current system.
- Block user access during the update.
- Keep instructions ready during installation.
- Apply the change and fully test the system.
- Restore system access (or go back to the old version if needed).
- Let staff know the change is finished.
- Log the change in the change management system.

- **Reversing a software upgrade:**
 - Software updates might cause unexpected issues.
 - Always be ready to undo an upgrade if needed.
- **Backleveling:**
 - Means switching back to an earlier software version after trying an upgrade.
 - There are no strict rules—each case may differ.
 - Always follow the vendor's documentation.
 - For network operating systems (NOS), ask for advice from other professionals too.

Physical Security Controls

- **Limit physical access to important equipment:**
 - Only trusted network staff should be allowed entry.
 - Security policies must clearly define who can access what.
- **Advanced door access systems:**
 - **Keypad or cipher locks:** These require a code to unlock the door and can be physical or electronic.
 - **Access badges:** Can include proximity cards, passive cards, or active cards for identity verification.
- **Advanced door access controls (continued):**
 - **Biometrics:** Scans a person's unique physical features.
 - **Mantraps:** Two-door systems where the first door must close before the second opens, used to control entry.
- **AIT (Advanced Imaging Technology) machines:**
 - Use millimeter-wave scanners to show cartoon-like images and highlight any suspicious areas for security staff.
- IT departments often use **CCTV systems** to monitor secure rooms.
- **IP cameras** are commonly placed in **data centers**.
- A **central security office** may:
 - Show multiple camera feeds at the same time, or
 - Automatically switch between different camera views.
- **Which rooms contain critical systems or data and must be secured?**
 - Server rooms and data storage areas.
- **Through what means might intruders gain access to the facility, computer room, data room, network closet, or data storage areas?**
 - Through unlocked doors or stolen access cards.
- **How and to what extent are authorized personnel granted entry?**
 - With ID cards or codes, only to certain areas.
- **Are employees instructed to ensure security after entering or leaving secured areas?**
 - Yes, they must close doors and report anything unusual.

- **Are authentication methods difficult to forge or circumvent?**
→ Yes, strong methods like biometrics or smart cards are used.
 - **Do supervisors or security personnel make periodic physical security checks?**
→ Yes, regular checks are done to keep areas secure.
 - **Are all combinations, codes, or other access means to computer facilities protected at all times, and are those combinations changed frequently?**
→ Yes, they are protected and changed often.
 - **Do you have a plan for documenting and responding to physical security breaches?**
→ Yes, there is a response plan for such situations.
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- To protect data on old or unused hard drives:
 - Use a **drive sanitizer program** to make sure files can't be recovered.
 - **Degausser:**
 - A device that erases hard drives using magnetic force.